Reliable Multimodal AI for Chest X-ray Report Generation and QA

CXR-Intelligence | Technical case study | November 6, 2026

Abstract

CXR-Intelligence implements chest X-ray report generation and retrieval-augmented question answering with MedGemma, BiomedCLIP and ColPali components. This case study reports an engineering audit that separates mock execution from real-model evidence, rejects invalid evaluation inputs, and removes unsupported performance conclusions. Software tests and synthetic smoke execution were verified; clinical accuracy and comparative real-model performance remain unevaluated in this audit.

Problem and research questions

Retrieved reports can contain findings unsupported by the query image. The proposed research asks whether visual retrieval improves report grounding, how different retrievers compare under controlled conditions, and when negation or laterality errors occur. These are research questions, not established findings.

Implemented system and contribution scope

The code provides an image-to-report pathway and an image/question-to-retrieval-to- answer pathway. BiomedCLIP and ColPali wrappers build visual indexes; MedGemma generates reports or answers using the image and optional context. Evaluation entry points coordinate inference and text-similarity scoring. A Gradio interface supports demonstrations.

This audit adds shared input validation, provenance checks, truthful metric naming, mode-separated outputs, regression tests and evidence-focused documentation. The repository does not establish who authored each original component. Personal/team contribution attribution must be confirmed before using this study as a first-person application statement.

Data and method

No patient dataset was processed in this audit. The software smoke test uses ten synthetic image/report fixtures and ten synthetic QA fixtures, with mock models. These fixtures test execution and serialization, not image understanding.

For future real runs, metadata must provide actual images, reference reports, study IDs and explicit splits. Patient IDs are required for RAG provenance. The loader checks study/patient overlap across metadata splits; the evaluator checks that index entries are training-only, real and disjoint from evaluation patients. Missing images, labels, requested splits, checkpoints or indexes fail explicitly. The checks rely on accurate supplied metadata and do not independently establish dataset authorization.

Verified results

Check	Observed outcome	Interpretation
Automated tests	76 passed; 26 new integrity tests	Engineering verification
Mock report execution	10/10 outputs	Functional smoke test
Mock QA execution	10/10 outputs	Functional smoke test
Real-model comparison	Not run	No performance conclusion
Clinical validation	Not performed	Not suitable for diagnostic claims

The mock lexical scores are retained in machine-readable evidence but are not presented as model accuracy. BERTScore and RadGraph were deliberately omitted from mock runs. Failed optional metrics use null and a status rather than zero. The separately named Clinical entity-overlap heuristic F1 does not reproduce RadGraph or validate negation.

Failure analysis and limitations

The earlier comparison path could supply missing images, create synthetic inputs without a mock flag, and generate conclusions independently of measured results. The QA path could replace unavailable images with blank ones. Historical metric failures could appear as zero, and a keyword heuristic carried the RadGraph label. Regression tests now exercise these failure conditions.

No evidence here establishes retriever superiority, fine-tuning gains, hallucination reduction or clinical safety. Legacy retrieval aggregates remain unverified. Patient- level excerpts and identifiers were removed from the current result files, but original content remains in Git history. Data authorization, model revision pinning, official split provenance, annotation quality and clinical review remain unresolved prerequisites.

Next experiment

Evaluate a permitted patient-disjoint sample with fixed generation settings: MedGemma zero-shot versus retrieval-augmented generation. Keep cases, prompts and retrieval depth comparable; document all deviations. Manually review supported/unsupported findings, negation and laterality using a defined protocol, and distinguish language similarity from grounding. Without qualified clinical review, call it report-grounding evaluation.

Reproducibility and references

python -m pytest -q

python scripts/run_comparison.py --mock --n-samples 10

Evidence: results/audit_2026_11_06/; source baseline: 13374a42fd536c22317d5aca434e19fc74892b5e plus the audit changes.

Repository: https://github.com/Roaa-838/cxr-intelligence2

Dataset reference: Johnson et al., MIMIC-CXR v2.1.0, PhysioNet (2024), https://physionet.org/content/mimic-cxr/2.1.0/ . Dataset use is subject to its access terms. This case study is a technical report, not a peer-reviewed publication.